Extra Bite #6: Amplifier Classes and Transistor Stages

A, AB, C, D -- how much of the cycle the device conducts, and why it matters.

Group E7B · 12 pool questions · Standalone · 5-10 minute presentation · Practice exams at eham.net/exams

The Big Idea: Conduction Angle

Amplifier "classes" describe **how much of each input cycle the active device conducts**. Less conduction = higher efficiency but more distortion (and more harmonics that need filtering). More conduction = less distortion but more wasted heat.

Class	Each device conducts for	Typical use	Why pick it
A	360° (the entire cycle)	Low-distortion linear stages	Cleanest signal; bias point is **halfway between saturation and cutoff**. (E7B04) Worst efficiency.
AB push-pull	**More than 180° but less than 360°** per device (E7B01)	Linear power amps (SSB, AM, FM)	Two devices each take more than half the cycle and overlap a bit, so the output stays linear without each device dissipating a full cycle's worth.
C	Less than 180°	FM, CW, and frequency multipliers	Very efficient, but very nonlinear. **Class C on an SSB phone signal causes signal distortion and excessive bandwidth.** (E7B07)
D (switching)	Output device is either fully on or fully off, switched at the signal rate	High-efficiency RF and audio amps	**Uses switching technology to achieve high efficiency.** (E7B02) Device is at **saturation or cutoff most of the time**. (E7B08)

Class D requires a filter at the output to remove harmonic content (E7B03). Without it the square switching waveform sprays harmonics across the spectrum.

Transistor Stage Configurations

Configuration	Characteristic	Q
Common emitter	Voltage and current gain; phase inverts input to output. The workhorse small-signal stage.	E7B12
Emitter follower (common collector)	Voltage gain 1, current gain high. **Input and output signals in phase.** Low output impedance.	E7B09
Grounded-grid (tube) / **common base** (transistor)	**Low input impedance** – characteristic of grounded-grid amps. Often used as the final RF stage in tube linears.	E7B06

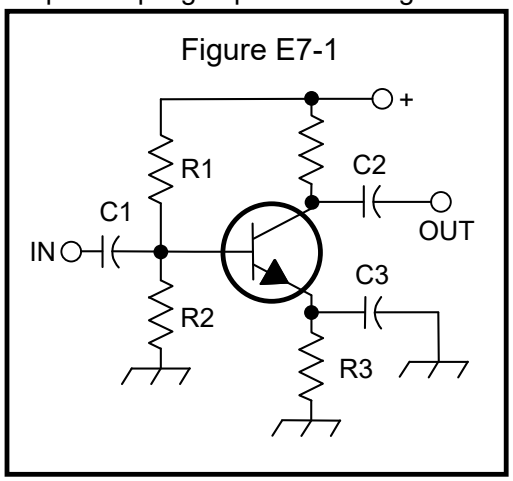
Parasitic Oscillation

Real RF power amplifiers can break into unwanted oscillations at frequencies far from the operating frequency, driven by stray inductance and capacitance. **Install parasitic suppressors and/or neutralize the stage.** (E7B05) Parasitic suppressors are small RLC networks in the plate or output lead that kill gain at the parasitic frequency; neutralization uses a feedback cap to cancel the path that lets the oscillation form.

Reading Figure E7-1 (Official NCVEC Diagram)

Three exam questions (E7B10, E7B11, E7B12) refer to this figure. It is a single-transistor amplifier with input and output coupling caps and biasing resistors.

Figure E7-1



Label	What it does	Q
Circuit type	Emitter at AC ground (through C3), input drives the base, output from collector → **common-emitter** amplifier. (E7B12)	E7B12
R1, R2	**Voltage divider bias.** Set the DC base voltage to a stable point. (E7B10)	E7B10
R3	**Self bias.** Emitter resistor; rising current raises emitter, reduces V_{BE}, pulls current back. (E7B11)	E7B11
C1, C2	Input/output AC-coupling caps (block DC, pass signal).	—
C3	Emitter-bypass cap: shorts R3 at signal frequencies to preserve AC gain.	—

Mental Cheat Sheet

- Class A: 360° conduction; bias **halfway between saturation and cutoff**.
- Class AB push-pull: each device more than 180° but less than 360°.
- Class C: under 180°; never use on SSB (distortion + excessive bandwidth).
- Class D: switching; output filter required to kill harmonics.
- Switching amps efficient because device is at **saturation or cutoff most of the time**.
- Emitter follower (common collector): in-phase, low Z out.
- Grounded-grid amp: **low input impedance**.
- Parasitic oscillation fix: **parasitic suppressors and/or neutralize**.
- Figure E7-1 = common-emitter. R1/R2 = voltage divider bias. R3 = self bias.

Common Traps

- Class AB push-pull is “more than 180 but less than 360” PER DEVICE, not the whole amp. Two devices share the full cycle.
- Class C on SSB causes **distortion AND excessive bandwidth**; “reduced intermod” is a trap.
- Class D efficiency comes from **saturation/cutoff operation**, not “higher voltages” or “differential inputs.”
- Emitter follower is **in phase**, NOT phase-inverted (that is common emitter).
- Grounded-grid amp is **low** input impedance, not high.
- R1/R2 in Figure E7-1 are **voltage divider** bias; “self bias” is R3 alone.

Takeaway: Amplifier class is conduction angle – less conduction means more efficiency and more distortion. Match the class to the signal: A or AB for linear (SSB, AM), C for constant-envelope (FM, CW, multipliers), D for switching. The configurations (common-emitter, emitter follower, grounded-grid) describe which pin is grounded for AC and define the impedance/phase behavior.

Bite 6 Question Bank (E7B)

12 questions. Read, decide, check the answer key at the end.

E7B01. For what portion of the signal cycle does each active element in a push-pull, Class AB amplifier conduct?
A. More than 180 degrees but less than 360 degrees.
B. Exactly 180 degrees.
C. The entire cycle.
D. Less than 180 degrees.

E7B02. What is a Class D amplifier?
A. An amplifier that uses switching technology to achieve high efficiency.
B. A low power amplifier that uses a differential amplifier for improved linearity.
C. An amplifier that uses drift-mode FETs for high efficiency.
D. An amplifier biased to be relatively free from distortion.

E7B03. What circuit is required at the output of an RF switching amplifier?
A. A filter to remove harmonic content.
B. A high-pass filter to compensate for low gain at low frequencies.
C. A matched load resistor to prevent damage by switching transients.
D. A temperature compensating load resistor to improve linearity.

E7B04. What is the operating point of a Class A common emitter amplifier?
A. Approximately halfway between saturation and cutoff.
B. Approximately halfway between the emitter voltage and the base voltage.
C. At a point where the bias resistor equals the load resistor.
D. At a point where the load line intersects the zero bias current curve.

E7B05. What can be done to prevent unwanted oscillations in an RF power amplifier?
A. Tune the stage for minimum loading.
B. Tune both the input and output for maximum power.
C. Install parasitic suppressors and/or neutralize the stage.
D. Use a phase inverter in the output filter.

E7B06. What is a characteristic of a grounded-grid amplifier?
A. High power gain.
B. Low input impedance.
C. High electrostatic damage protection.
D. Low bandwidth.

E7B07. Which of the following is the likely result of using a Class C amplifier to amplify a single-sideband phone signal?
A. Reduced intermodulation products.
B. Increased overall intelligibility.
C. Reduced third-order intermodulation.
D. Signal distortion and excessive bandwidth.

E7B08. Why are switching amplifiers more efficient than linear amplifiers?
A. Switching amplifiers operate at higher voltages.
B. The switching device is at saturation or cutoff most of the time.
C. Linear amplifiers have high gain resulting in higher harmonic content.
D. Switching amplifiers use push-pull circuits.

E7B09. What is characteristic of an emitter follower (or common collector) amplifier?
A. Low input impedance and phase inversion from input to output.
B. Differential inputs and single output.
C. Acts as an OR circuit if one input is grounded.
D. Input and output signals in-phase.

E7B10. In Figure E7-1, what is the purpose of R1 and R2?
A. Load resistors.
B. Voltage divider bias.
C. Self bias.
D. Feedback.

E7B11. In Figure E7-1, what is the purpose of R3?
A. Fixed bias.
B. Emitter bypass.
C. Output load resistor.
D. Self bias.

E7B12. What type of amplifier circuit is shown in Figure E7-1?
A. Common base.
B. Common collector.
C. Common emitter.
D. Emitter follower.

Answer Key

E7B01: A	E7B02: A	E7B03: A	E7B04: A
E7B05: C	E7B06: B	E7B07: D	E7B08: B
E7B09: D	E7B10: B	E7B11: D	E7B12: C